Group-Project SSBI

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Benchmark of comparative single-cell analysis approaches

The Data

from Horowitz et al. (2013)

Provided donors	20
CMV+/CMV-	9/11
Analysis markers	37
Available gated NK cells	261,593
Balanced analysis sample	40,000

The context

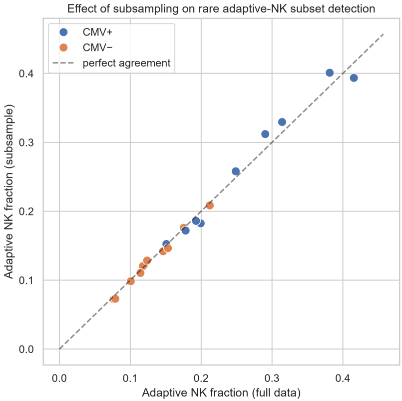
Mass cytometry (CyTOF): simultaneous protein-marker measurements on individual cells.
Natural Killer (NK) cells: immune cells with a high cell diversity

From some donors a subpopulation of NK cells shows a characteristic expression of proteins typical after a CMV infection, which was given in the data

CMV labels describe donors; they do not identify infected individual cells.

Horowitz et al., Sci. Transl. Med. 2013; Arvaniti & Claassen, Nat. Commun. 2017. Provided 20-donor subset

Does the balanced subsample preserve the selected phenotype?

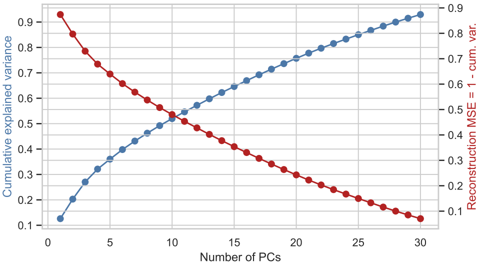


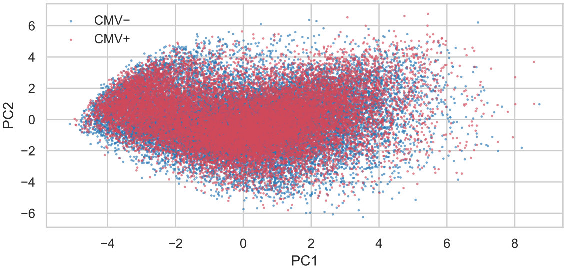
Compare randomly choosen **2,000 cells per donor** with all available gated NK cells from that donor.

NKG2C$^+$/CD57$^+$ is a marker phenotype associated with **memory-like ("adaptive") NK cells** after CMV exposure.

Positivity: two-component GMM crossover per marker on all 261,593 cells. This check concerns one approximate phenotype, not every rare population.

PCA



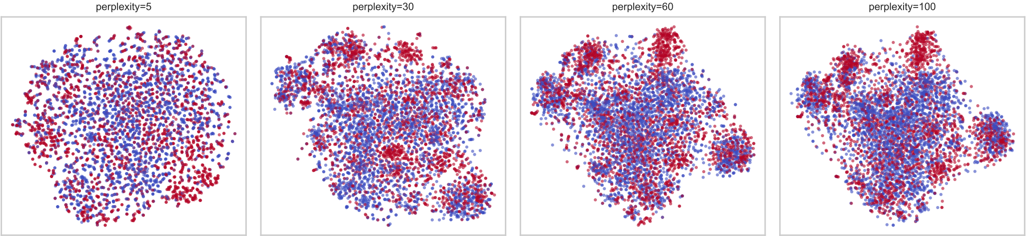


Retention choice: 29 PCs reach the 90% threshold and become the input to t-SNE/UMAP.

29 PCs retain 91.50%, the plotted two retain only 20.29%

40,000 cells, arcsinh + z-scoring. Spectrum verified from the same 37-marker covariance matrix

t-SNE: changing perplexity changes the neighbourhood scale



Input: 5,000 cells and 29 PCs reduce sweep time and memory.

The top-scoring of 5, 15, 30, 40, 50, 60, 70 and 100. **Selected: 60**

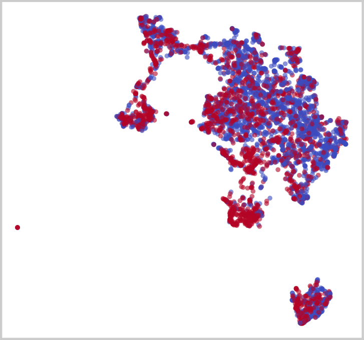
Perplexity	Trustworthiness
50	0.92597
60	0.92605
100	0.92580

T: penalises false low-dimensional neighbours by their high-dimensional ranks.

Selected panels; colours indicate donor CMV status.

UMAP: neighbourhood scale versus point packing

nn=5, md=0.0

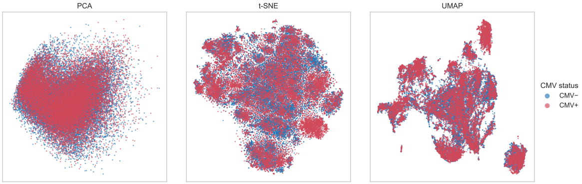


$n_{\text{neighbors}}$	min_dist	$T(15)$
5	0.0	0.890139
5	0.1	0.879145
30	0.0	0.877151

Selected: **5 neighbours, min_dist = 0.0**.

Four corner settings from the 16-panel saved grid, colours indicate CMV status. Top three rows ranked by saved Trustworthiness on the same 5,000 cells and 29 PCs.

Local preservation and global distances favour different maps



Embedding	$T(15)\uparrow$	$R(15)\uparrow$	$r_d\uparrow$	$S_{\mathrm{CMV}}\uparrow$
PCA	0.7258	0.0072	**0.6204**	0.0016
t-SNE	**0.9612**	**0.2232**	0.4868	0.0104
UMAP	0.9009	0.1126	0.3954	**0.0178**

Local structure

T: **Trustworthiness** – false neighbours, ranked.

R: **kNN preservation** – exact overlap; small values are normal at $n = 40{,}000$ (chance level ≈ 0.0004).

t-SNE wins both: it directly optimises neighbour probabilities.

Global / label structure

r_d: **Pearson correlation** of all pairwise distances.
PCA wins: a linear projection built to preserve overall variance/distance.

S_{CMV}: **Silhouette score** - separation by donor CMV label – near zero for all three maps, as expected: only a minority subset of each CMV+ donor's cells is actually altered.

t-SNE leads locally; PCA leads in distance correlation.

40,000 cells; 29 PCs; k=15. Distance correlation: fixed 2,000-cell sample.

Which maps are most similar?

Pair	$R(15)\uparrow$	$M^2\downarrow$
PCA vs t-SNE	0.00585	**0.45189**
PCA vs UMAP	0.00410	0.73390
t-SNE vs UMAP	**0.12532**	0.72369

R values look small in absolute terms, but sit far above the ~ 0.0004 level expected by chance at $n = 40{,}000$, $k = 15$.

M^2 – Procrustes disparity

Align two maps by translation, rotation/reflection and scaling; measure the remaining mismatch. Lower = closer overall shape after alignment.

Local neighbours (R): t-SNE and UMAP are the most similar pair.

Aligned global shape (M^2): PCA and t-SNE are the most similar pair.

R and M^2 disagree because they ask different questions: exact shared neighbours vs. overall shape after optimal alignment

All three pairs evaluated on the same 40,000 cells in the same order. R(15) corrected from saved coordinates; SciPy Procrustes values independently reproduced. Neither metric identifies biologically correct cell types.

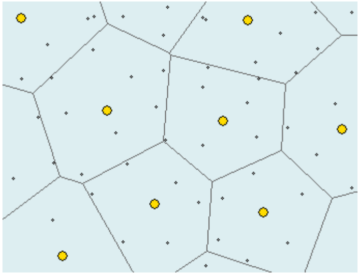
K-means: nearest centres and Voronoi regions

Choose K, then repeat

1. Assign every cell to its nearest centre.
2. Update each centre to its cluster mean.

K: number of clusters. Each cell gets one label.

Geometric model
Convex regions with linear boundaries.
Probabilistic model: hard-assignment Gaussian clusters with a shared spherical covariance and equal prior weights.



Yellow points: centres. Each region contains points closest to its centre.

Our data: 20,000 live PBMCs, 37 markers, 20 donors.

Which choice of K resolves meaningful populations, including potentially rare subsets?

Figure excerpt: Ulrike von Luxburg, Statistical Machine Learning, Summer 2024, slide 840 (PDF p. 844), cropped. Gaussian interpretation: Macke / Hennig, Lecture 11.

Silhouette and Davies–Bouldin: what gets averaged?

Silhouette: cell-weighted average

$$\bar{s} = \sum_{c=1}^{K} \frac{n_c}{N}\, \bar{s}_c$$

n_c: number of cells in cluster c.
N: total number of cells.
$\bar{s}_c$: mean silhouette in cluster c.

A group with **1 % of cells** has **1 % weight** for its own mean silhouette.

DB: equal weight per cluster

$$DB = \frac{1}{K} \sum_{c=1}^{K} \max_{j \neq c} \frac{S_c + S_j}{\|\mu_c - \mu_j\|}$$

S_c: mean distance of cells to their centre.
μ_c: centre (mean vector) of cluster c.

Each cluster has $1/K$ **weight**. Lower is better.

DB gives every cluster same weight.

Definitions: scikit-learn clustering documentation.

Subsampling within the observed cells

Why keep the initial sample?

- A fresh sample may omit a rare population that we observed initially.
- Different cells also prevent a direct membership comparison.
- We first test how well observed groups survive removing cells.

Our stability experiment

1. Keep 80 % of cells within each donor, without replacement.
2. Refit scaling and clustering with the same parameter settings.
3. Compare both partitions on the **same retained cell IDs**.

Ten repeats, 16,000 cells each, with changing random seeds.

This measures stability conditional on our initial sample. Even 80 % subsampling can lose a tiny group.

Our notebook implementation.

Stability: global ARI or cluster-wise Jaccard?

ARI compares entire partitions

It measures agreement over cell pairs and adjusts for chance.

A group of n cells contains $n(n-1)/2$ within-group pairs. Large groups therefore contribute far more pairs.

Splitting a rare group changes relatively few pairs, so global ARI can remain high.

Jaccard follows each population

$$J(C, D) = \frac{|C \cap D|}{|C \cup D|}$$

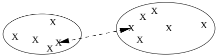
C: reference cluster among the retained cells.
D: cluster from the new fit on those same cells.

Our choice: Jaccard per cluster, then equal cluster weights. Both approaches work without ground truth.

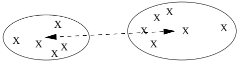
SOURCE

Agglomerative clustering: which groups merge?

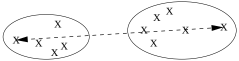
Single linkage: $dist(C, C') = \min_{x \in C, y \in C'} d(x, y)$



Average linkage: $dist(C, C') = \frac{\sum_{x \in C, y \in C'} d(x,y)}{|C| \cdot |C'|}$



Complete linkage: $dist(C, C') = \max_{x \in C, y \in C'} d(x, y)$



C, C': two clusters; x, y: cells.
$d(x, y)$: Euclidean distance; $|C|$: cell count.

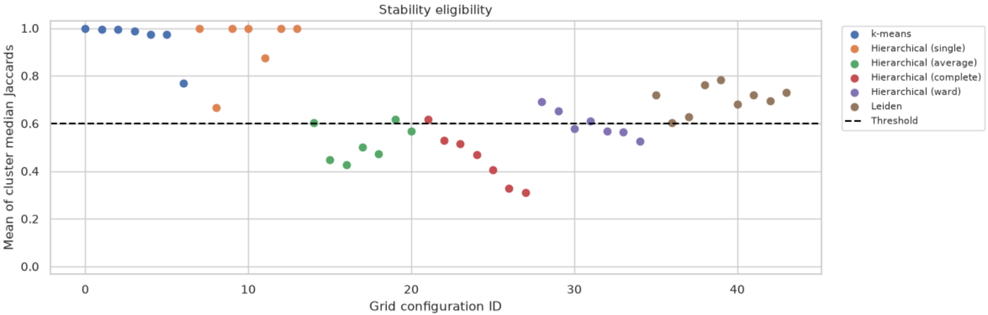
Start with one cluster per cell
Merge the closest groups repeatedly. Cut the resulting tree at the desired cluster count.

Ward uses a different merge rule
Choose the smallest increase in within-cluster squared error (very k-means like). This favours compact groups.

Earlier merges remain fixed in every variant.

Figure excerpt: Ulrike von Luxburg, Statistical Machine Learning, Summer 2024, slide 850 (PDF p. 854), cropped. Ward: SciPy linkage documentation.

Selection: stability first, then Davies–Bouldin



Each dot: one configuration. Keep mean stability ≥ 0.6 (arbitrary), then minimise DB per method.

K-means++ and 10 starts
Reduce sensitivity to poor initial centres.
Stochastic **farthest first** heuristic

Single linkage and chaining
A large group can grow through large chain of local neighbourhoods

Original saved notebook plots.

Excellent scores can describe an unhelpful partition

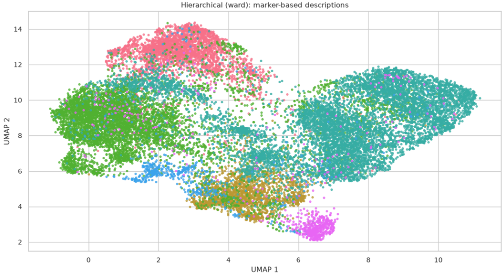
Method	Parameters	Clusters	DB ↓	Stability	Largest cluster
K-means	$k = 4$	4	2.363	0.995	37.780 %
Single	$k = 2$	2	0.264	1.000	99.995 %
Average	$k = 2$	2	0.894	0.602	99.975 %
Complete	$k = 2$	2	0.948	0.617	99.960 %
Ward	$k = 6$	6	2.522	0.611	38.220 %
Leiden	$n = 30, r = 1.2$	18	2.361	0.683	17.940 %

Single linkage: 19,999 cells + 1 cell
DB = 0.264 and mean stability = 1.

An isolated event can score well and recur. These metrics alone do not establish a rare cell population.

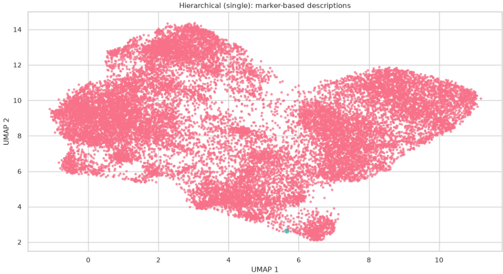
Ward and single linkage on the same UMAP

Ward: 6 clusters



DB: **2.522** Mean stability: **0.611**

Single: 19,999 cells + 1 cell



DB: **0.264** Mean stability: **1.000**

Ward divides the broad cloud. Single linkage assigns almost the entire cloud to one cluster.

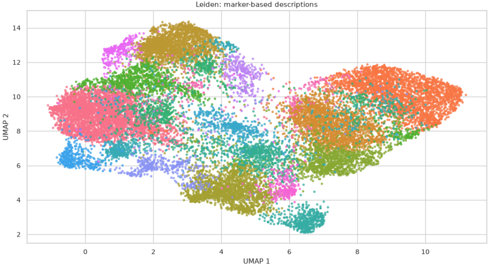
Original saved notebook plots.

K-means and Leiden give different resolutions

K-means: 4 clusters

3,148–7,556 cells per cluster
DB: **2.363** Mean stability: **0.995**

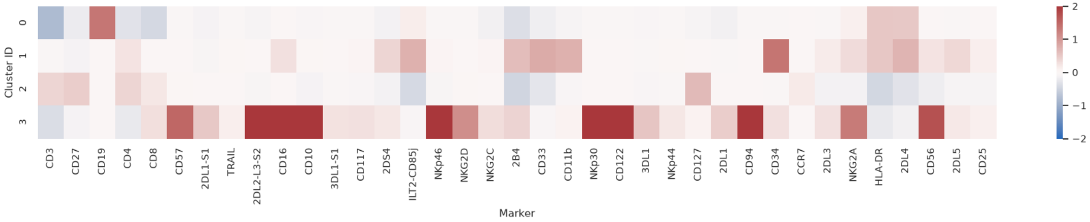
Leiden: 18 clusters



223–3,588 cells per cluster
DB: **2.361** Mean stability: **0.683**

Original notebook plots on the same UMAP coordinates.

K-means: marker profiles of the four groups



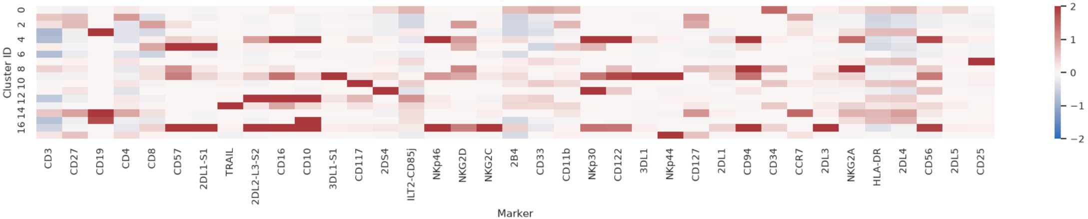
$$\text{Relative marker score} = \frac{\text{cluster median} - \text{sample median}}{\text{sample IQR}} \quad \text{vs.} \quad \frac{\text{cluster mean} - \text{sample mean}}{\text{sample standard deviation}}$$

For each marker after arcsinh: $\text{IQR} = Q_{75\%} - Q_{25\%}$. Red: higher median; blue: lower.

more robust statistical marker

Original notebook panel. NIST

Leiden: marker profiles of the finer groups



Original notebook panel.

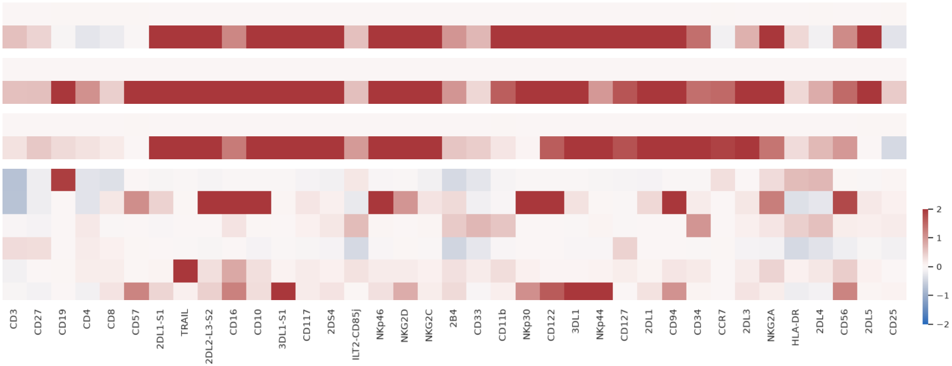
Hierarchical clustering: marker profiles

Single
C0, C1

Average
C0, C1

Complete
C0, C1

Ward
C0–C5



Single / average / complete: tiny groups with broadly elevated markers. Ward: more differentiated profiles.

Original notebook heatmap strips.

Assessment for the cell-population question

K-means: broad, reproducible groups
4 clusters, mean stability 0.995.

Ward: intermediate resolution
6 clusters with differentiated marker profiles.
Mean stability 0.611; 3 of 6 groups below 0.6.

Leiden: finer groups
18 clusters, mean stability 0.683.
More detail, but 4 of 18 groups below 0.6.

Single, average and complete
Selected solutions mainly separate a few events from almost all other cells.

Rare cell populations
Equal cluster weighting helps evaluate small groups. A biological interpretation also needs coherent markers and replication.

K-means provides a stable baseline. Ward and Leiden add detail with weaker reproducibility.

Our saved results for 20,000 live PBMCs.

From cells to donor-level prediction

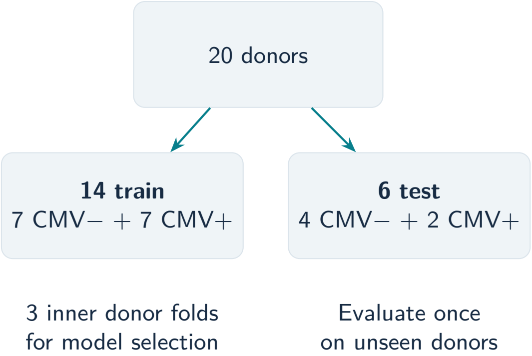
One CMV label per donor

Donors	20
CMV− / +	11 / 9
Protein markers	37
Input gate	`gated_alive`

$$a_{ij} = \operatorname{asinh}(x_{ij}/5)$$

i: cell; j: marker; x_{ij}: raw value; a_{ij}: transformed value; 5: cofactor.

CellCNN/SVM: fit marker scaling on training donors only. Citrus uses ArcSinh values directly.



Horowitz et al. (2013), doi:10.1126/scitranslmed.3006702. Arvaniti & Claassen (2017), doi:10.1038/ncomms14825.

CellCNN learns filters for informative cells

Cell group 3,000 × 37 → **3–5 filters** linear + ReLU → **Top 1%** mean per filter → **Softmax** CMV probability

Learn which cells matter

$$r_{if} = \max(0, w_f^\top z_i + b_f)$$

$\boldsymbol{i}$: cell; $\boldsymbol{f}$: filter; $\boldsymbol{z_i}$: 37 standardized markers; $\boldsymbol{w_f}$: filter weights; $\boldsymbol{b_f}$: bias; $\boldsymbol{r_{if}}$: response. $\boldsymbol{w_f^\top z_i}$: weighted sum; $\boldsymbol{\max(0, \cdot)}$: ReLU.

A filter scores a marker combination. Pooling focuses on the strongest responses.

Our paper-guided implementation

- PyTorch; 200 cell groups per training donor.
- Choose the best inner-fold network.
- Adaptations: fixed 3/4/5-filter search; average five test-input probabilities.

Arvaniti & Claassen (2017), doi:10.1038/ncomms14825.

Running the original Citrus implementation

1. R environment
separate Conda env

→

2. Original packages
fixed Git commits

→

3. Notebook kernel
shared donor splits

R 4.5, `flowCore` and `glmnet`; C++ compiler and `make` supplied by Conda.

Citrus 0.8 and `Rclusterpp` installed from the official repositories at pinned commits.

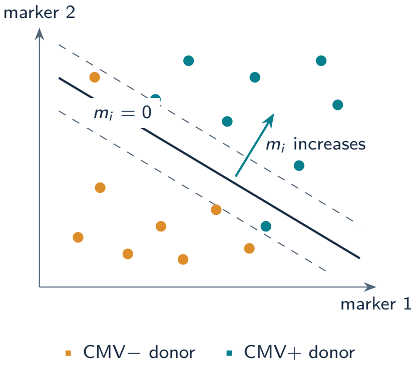
Register `IRkernel`; the R notebook connects the original functions to our fixed donor folds.

Two routes to a paper-guided implementation

CellCNN: new PyTorch implementation based on the paper and reference code.

Citrus: original legacy package code; recorded runtime: R 4.5.3.

A linear SVM assigns a score to each cell



Train on donor-labelled cells

10,000 cells per training donor; each cell receives its donor's label.

$$m_i = w^\top z_i + b$$

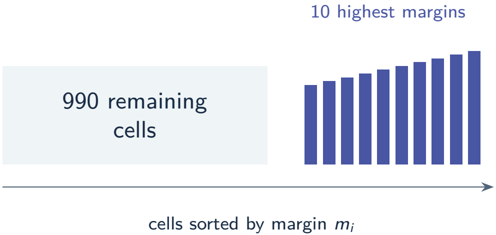
A larger margin means stronger evidence in the model's positive direction.

Linear SVM: L2 penalty and squared-hinge loss. The margin is a score, not a probability.

Arvaniti & Claassen (2017), doi:10.1038/ncomms14825.

Our adaptation: aggregate the highest cell margins

A few high-scoring cells can matter



Toy example: 1,000 cells; keep 10.
Margins 2.1, 2.2, . . . , 3.0 give $s_d = 2.55$.
At $\tau = 1.80$: predict CMV+.

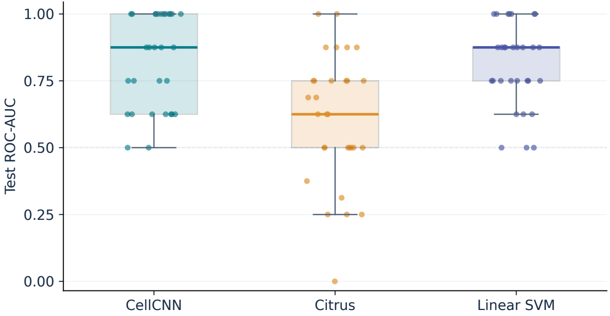
One score per donor

$$k_d = \lceil 0.01 N_d \rceil, \qquad s_d = \frac{1}{k_d} \sum_{i \in I_d} m_i$$

$\boldsymbol{d}$: donor; $\boldsymbol{N_d}$: its cell count; $\boldsymbol{k_d}$: count rounded up; $\boldsymbol{I_d}$: its top-scoring cell indices; $\boldsymbol{i}$: cell; $\boldsymbol{m_i}$: cell margin; $\boldsymbol{s_d}$: mean donor score; $\boldsymbol{\tau}$: donor threshold.

1. Select loss weight $C \in \{0.01, 0.1, 1\}$ by inner donor ROC-AUC.
2. Learn τ from inner out-of-fold donor scores.
3. Refit on 14 training donors; score all test cells.

Performance across 30 shared donor splits



Six test donors per split. Boxes show Q1–Q3; dots show individual splits.

BA thresholds: 0.5 for CellCNN/Citrus; inner-trained for SVM.

Median across 30 splits

Method	AUC	AP	BA
CellCNN	0.875	0.833	0.625
Citrus	0.625	0.542	0.500
Linear SVM	0.875	0.833	0.625

AUC: ranking of CMV+ versus CMV−.
AP: average precision.
BA: balanced accuracy.

CellCNN: select cells with a strong filter response

Cell response
apply a learned filter

→

Normalize
training-reference max

→

Select cells
above half-maximum

One score per cell and filter

$$q_{if} = \frac{r_{if}}{M_f}, \qquad q_{if} > 0.5$$

i: cell
f: filter
$\boldsymbol{r_{if}}$: filter response
$\boldsymbol{M_f}$: its positive maximum on the original training reference
$\boldsymbol{q_{if}}$: normalized score.

A simple example

Reference maximum: **10**. Selection threshold: **5**.

Responses **8 and 6** pass; responses **5 and 2** do not.

Use effective filters. Their output weights determine the CMV direction.

Citrus: select clusters used by the classifier

Cluster frequency
one feature per donor

→

Model coefficient
effect on CMV prediction

→

Selected cluster
saved mean marker profile

A score for a whole cluster

$$q_j = |\beta_j|, \qquad q_j > 10^{-10}$$

$\boldsymbol{j}$: cluster
$\boldsymbol{\beta_j}$: its frequency coefficient in the selected logistic model
$\boldsymbol{q_j}$: absolute coefficient.

The tolerance excludes numerical zeros.

What this selects

All clusters with an effective coefficient enter the interpretation.

Positive coefficient: higher frequency supports CMV+; negative: CMV−, with other features fixed.

The stored centroid summarizes the cluster's training cells.

SVM: select the strongest positive cell scores

Full test donor	→	Highest 1%	→	Positive selection
score every cell		rank within that donor		above donor threshold

1. Compute the cell score

$$m_i = w^\top z_i + b$$

$\boldsymbol{i}$: cell
$\boldsymbol{z_i}$: standardized marker vector
$\boldsymbol{w}$: learned weights
$\boldsymbol{b}$: intercept
$\boldsymbol{m_i}$: margin
$\boldsymbol{w^\top z_i}$: weighted sum.

2. Check the selected top cells

$$t_i = m_i - \tau, \qquad t_i > 0$$

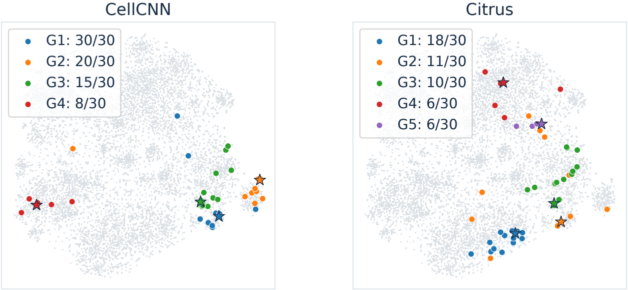
$\boldsymbol{\tau}$: donor threshold learned in inner validation
$\boldsymbol{t_i}$: cell score relative to that threshold.

A positive value passes **only within the highest 1%** of the full donor.

Which CellCNN and Citrus subsets recur?

Group similar subset centroids separately for each method; count distinct splits.

10,000 gated_alive cells; 500/donor; t-SNE perplexity 30; model splits 0–29.



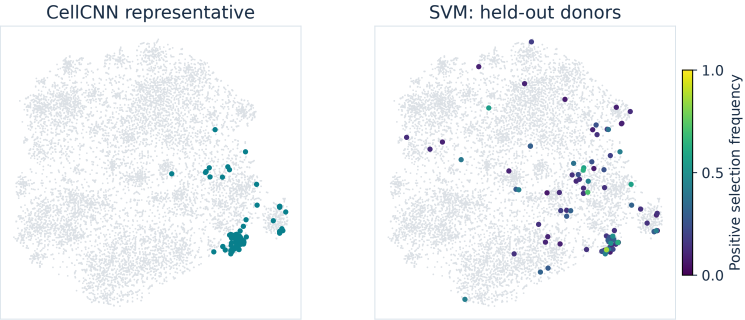
Show groups present in at least 6 of 30 splits (20%). Top groups: **30/30** and **18/30**, both positive.

Arvaniti & Claassen (2017), doi:10.1038/ncomms14825.

Which cells does the SVM repeatedly select?

Colour shows how often the same cell passes the positive SVM selection rule.

10,000 gated_alive cells; 500/donor; t-SNE perplexity 30; model splits 0–29.

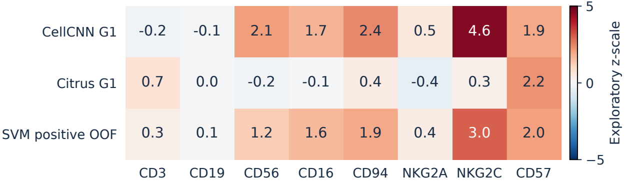


CellCNN: 87 map cells (exploratory). **SVM:** 93 map cells selected positively at least once.

Arvaniti & Claassen (2017), doi:10.1038/ncomms14825.

What do the marker profiles support?

CellCNN/Citrus: group representatives. SVM: positive test-cell profiles, with equal donor weights.



CellCNN

NKG2C/CD57 with CD56/CD16/CD94: compatible with a memory-like NK profile.

Citrus

Higher CD3/CD57 and much less NKG2C elevation suggest a T-associated pattern.

SVM positive OOF

NKG2C/CD57 and NK-associated markers are elevated; CD3 is also above the reference mean.

Arvaniti & Claassen (2017), doi:10.1038/ncomms14825. Horowitz et al. (2013), doi:10.1126/scitranslmed.3006702.

How should we score the learned populations?

Cluster geometry

Davies–Bouldin: compact, separated groups.

A distinct cell type need not distinguish CMV^+ from CMV^-.

Phenotype association

Does population frequency distinguish sample labels?

We have labels for samples, not individual cells.

We seek predictive, disease-associated cells, rather than a complete cell-type partition. Association does not establish causation.

Arvaniti & Claassen (2017); Davies & Bouldin (1979). Samples are represented as bags of cells.

Select a predictive filter, then count responding cells

1. Choose the filter with the strongest positive output contrast

$$\Delta_k = V_{1k} - V_{0k}, \qquad k^* = \arg\max_{k:\Delta_k>0} \Delta_k.$$

This selects by the learned classification weights, not the largest cell response.

2. Apply a half-maximum response cutoff

$$t_{k^*} = \tfrac{1}{2} \max_{x\in\text{Inner-Train}} h_{k^*}(x), \qquad f_b = \frac{1}{N_b}\sum_i \mathbf{1}\{h_{k^*}(x_{bi}) > t_{k^*}\}.$$

Calibrate on training cells; apply the same cutoff to every test donor.

Half-maximum is a heuristic, not a biological boundary. Its suitability must be checked and may require adaptation for our filters.

Existing selection and frequency rule: GrHa/06d. No positive output contrast: report population analysis as missing. Alternative cutoffs must be chosen without test labels.

References

Data and learning methods

Horowitz et al. (2013). Genetic and environmental determinants of human NK cell diversity revealed by mass cytometry. *Sci. Transl. Med.* 5:208ra145.

Arvaniti & Claassen (2017). Sensitive detection of rare disease-associated cell subsets via representation learning. *Nat. Commun.* 8:14825.

Bruggner et al. (2014). Citrus. *PNAS*.

van der Maaten & Hinton (2008). Visualizing data using t-SNE. *JMLR* 9:2579–2605.

McInnes, Healy & Melville (2018/2020). UMAP. arXiv:1802.03426.

Venna & Kaski (2001). Neighborhood preservation in nonlinear projection methods. *ICANN*, 485–491.

Clustering, lecture material and definitions

Traag, Waltman & van Eck (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Sci. Rep.* CC BY 4.0.

Ulrike von Luxburg (2024). *Statistical Machine Learning*, slides 840 and 850; cropped figures.

Jakob H. Macke / P. Hennig (2024). *Probabilistic Machine Learning*, Lecture 11, slide 15; adapted derivation, CC BY-NC-SA 3.0.

Official documentation: scikit-learn, SciPy, Scanpy / Leiden, UMAP.

NIST: interquartile range.

Project results: saved notebooks and exported tables; experiment settings and figure attributions accompany the slides.

Task 6: filter geometry and learnable pooling

Two extensions of our CellCNN baseline

Cell set $N \times p$ $\longrightarrow$ Mahalanobis-ReLU or Quadratic-ReLU $\longrightarrow$ Learnable Soft Top Fraction $\longrightarrow$ Sample label $\mathrm{CMV}^{+}/\mathrm{CMV}^{-}$

Which cell profiles respond? How broadly do we average?

Our baseline: linear ReLU filters with **fixed top-1 % pooling**.

Predict the sample phenotype from an unordered set of cells.

Arvaniti & Claassen (2017), Methods; project snapshot, 9 September 2026: 04c, GrHa/06d, GrHa-learnable-pooling/06b.

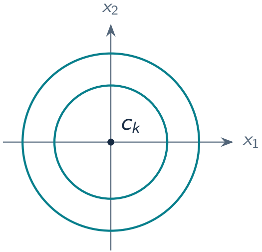
Starting point: an isotropic Gaussian filter

$$g_k(x) = \exp\left(-\frac{\|x - c_k\|^2}{2\ell_k^2}\right)$$

A smooth response around a prototype c_k, with one width ℓ_k for all marker directions.

- Equal deviations receive equal penalties in standardized coordinates.
- Standardization equalizes scale, not predictive relevance.
- One width cannot allow broad variation in one marker and narrow variation in another.

Equal-response contours



Spheres in p dimensions.

Isotropy is a useful baseline. We relax its equal-tolerance assumption with learned marker weights.

Gaussian radial basis function: Rasmussen & Williams (2006), GPML, Ch. 4. Model motivation, not an experimental finding.

Diagonal Mahalanobis distance

$$\boxed{d_k^2(x) = \frac{1}{p}\sum_{j=1}^{p} a_{kj}(x_j - c_{kj})^2 = \frac{1}{p}(x - c_k)^\top A_k(x - c_k)}$$

$$A_k = \mathrm{diag}(a_{k1}, \ldots, a_{kp}), \qquad a_{kj} > 0, \qquad \frac{1}{p}\sum_j a_{kj} = 1.$$

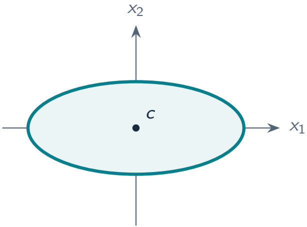
- c_k: preferred marker profile, learned through sample labels.
- a_{kj}: sensitivity to deviations along marker axis j.
- Larger a_{kj} means narrower tolerance, not higher expression.

Classical Mahalanobis uses $M = \Sigma^{-1}$. Here $M = A/p$ is learned discriminatively, rather than estimated as an inverse covariance.

Project definition: GrHa-learnable-pooling/06b. Positive parameterization and normalization: backup slide.

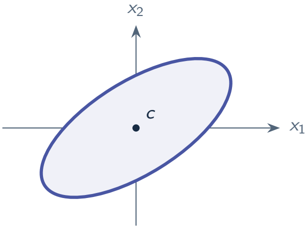
Diagonal or full Mahalanobis?

Diagonal: axis-aligned



$a_1\delta_1^2 + a_2\delta_2^2 = \text{constant}$

Full: rotation is possible



$a_1\delta_1^2 + 2a_{12}\delta_1\delta_2 + a_2\delta_2^2 = \text{constant}$

At $p = 37$: **37** diagonal versus **703** independent symmetric matrix entries per filter, before normalization; plus 37 center coordinates.

Markers can be coupled, but learning all cross-terms from only 20 donors risks fitting donor-specific geometry.

Own schematic contours; $703 = 37 \cdot 38/2$. Positive definite quadratic forms: Stanford EE263, Lecture 16.

Cell responses compared

Filter	Cell response $h_k(x)$	Preferred structure
Linear	$\mathrm{ReLU}(b_k + w_k^\top x)$	A direction in marker space
Mahalanobis	$\mathrm{ReLU}\Big(\rho_k - \frac{1}{p}(x - c_k)^\top A_k (x - c_k)\Big)$	Proximity to a center
Quadratic	$\mathrm{ReLU}\Big(b_k + w_k^\top x + \sum_j q_{kj} x_j^2\Big)$	Free curvature along each axis

Each filter applies the same parameters to every cell.

Quadratic uses $Q_k = \mathrm{diag}(q_{k1}, \ldots, q_{kp})$: no cross-terms, no rotated quadratic axes.

The preactivation defines the geometry; ReLU alone does not create a bounded cell population.

Project: GrHa/04c, 06b and 06d. Both target variants pool ReLU responses with Soft Top Fraction.

Mahalanobis is a restricted quadratic form

Expand the preactivation:

$$\rho - \frac{1}{p}(x-c)^\top A(x-c) = \underbrace{-\frac{1}{p}x^\top Ax}_{\text{quadratic}} + \underbrace{\frac{2}{p}c^\top Ax}_{\text{linear}} + \underbrace{\rho - \frac{1}{p}c^\top Ac}_{\text{constant}}.$$

Mahalanobis-ReLU

$q_j = -a_j/p < 0$, $w_j = 2a_jc_j/p$.
Negative curvature and normalized scale impose a bounded region around c.

Quadratic-ReLU

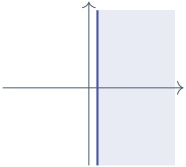
q_j, w_j and b are free parameters.
Open regions are possible; $q = 0$ recovers the linear filter.

The quadratic filter contains this Mahalanobis preactivation as a special case, without enforcing its geometry.

Own algebraic derivation; implementation: GrHa/06d. Here a filter is a cell-scoring function, not a pairwise Mercer kernel.

Which shapes can a quadratic filter detect?

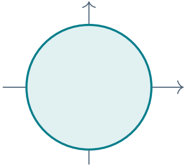
Linear



$z = x_1 - 0.2$

Half-space

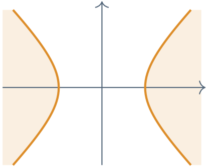
Negative curvature



$z = 2.1 - x_1^2 - x_2^2$

Inside a circle

Mixed signs



$z = x_1^2 - x_2^2 - 1$

Open regions

Shading: $z(x) > 0$, hence positive ReLU response.
Without cross-terms $x_i x_j$, the quadratic axes cannot rotate freely.

Quadratic can detect a circular region. Greater flexibility does not guarantee better prediction.

Own analytical examples, not cell measurements. Boundaries follow from the displayed polynomials.

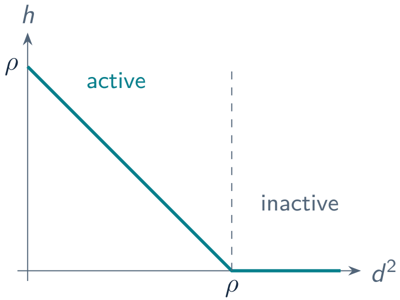
ReLU introduces an absolute activation boundary

$$h_k(x) = \mathrm{ReLU}(\rho_k - d_k^2(x))$$

$\rho_k > 0$ learned positive threshold

Inside $d_k^2 < \rho_k$: positive response.
Outside: zero response and zero local gradient through ReLU.

ρ_k is learned in *squared-distance units*. The corresponding metric radius is $\sqrt{\rho_k}$.



ReLU can suppress background responses, but a filter that is inactive on all training cells may receive no useful gradient.

GrHa/06b: ReLU + mean. ReLU + Soft Top Fraction is the target combination; the read learnable branch still uses $s = -d^2$.

Hard top-k pooling

Keep the k largest cell responses and calculate their mean.

$$s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(N)}, \qquad \boxed{P_{\text{top-}k} = \frac{1}{k}\sum_{r=1}^{k} s_{(r)}}$$

For a top fraction α, choose

$$k = \max(1, \lfloor \alpha N \rfloor). \qquad \text{Our baseline: } \alpha = 0.01.$$

Problem: the selection size is discrete.

Changing α either leaves k unchanged or adds a whole cell.

The hard selection is not differentiable at changes in k; between them, its gradient with respect to α is zero.

Project top-k mean pooling: GrHa/04c and 06d. For fixed k, selected response values still receive gradients.

Relax the hard selection

Hard top-k can be written as a binary mask.

$$m_r^{\mathrm{hard}} = \begin{cases} 1, & r \le k \\ 0, & r > k \end{cases}$$

Keep the first k sorted responses.

Replace the abrupt decision with a smooth transition.

$$\underbrace{m_r^{\mathrm{hard}} \in \{0,1\}}_{\text{keep or discard}} \longrightarrow \underbrace{m_r = \mathrm{sigmoid}\left(\frac{\alpha - u_r}{\tau}\right)}_{\text{smooth rank mask}}$$

u_r: relative rank position; α: fraction cutoff; τ: transition width.

The sigmoid gives high-response ranks values near one and lower ranks values near zero.

Smooth relaxation of rank-based selection. The complete pooling operator is defined on the next slide.

Soft Top Fraction: definition

For each filter, sort the cell responses:

$$s_{(1)} \geq \cdots \geq s_{(N)}.$$

$$\underbrace{u_r = \frac{r - \frac{1}{2}}{N}}_{\text{relative rank position}}, \qquad \underbrace{\alpha = \mathrm{sigmoid}(\theta)}_{\text{learned fraction}}$$

$$m_r = \mathrm{sigmoid}\left(\frac{\alpha - u_r}{\tau}\right), \qquad \boxed{P_{\mathrm{soft}} = \frac{\sum_{r=1}^{N} m_r s_{(r)}}{\sum_{r=1}^{N} m_r}}$$

Dividing by the mask sum gives a weighted mean. One α is learned per filter and shared across samples; τ stays fixed.

The weights are differentiable in α, allowing the classification loss to learn the pooling fraction.

GrHa-learnable-pooling/06b: initial $\alpha = 0.01$, fixed $\tau = 0.002$. Sorting and ReLU are differentiable almost everywhere; derivation in backup.

Regularization: different reference models

Quadratic

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + 10^{-4}(\|W\|_F^2 + \|Q\|_F^2 + \|V\|_F^2)$$

$Q = (q_{kj})$: quadratic coefficient table.

Initialization at $q = 0$ reproduces the linear baseline. L2 on Q favors little additional curvature.

Mahalanobis, learnable branch

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + 10^{-4}\|V\|_F^2 + 10^{-3}\,\mathrm{mean}_{k,j}(a_{kj} - 1)^2$$

$\mathrm{mean}_j\, a_{kj} = 1$ fixes the scale.

The geometry penalty favors equal axis weights. In the read code, it is applied only to the training loss.

With normalized a, $\mathrm{mean}(a^2) = 1 + \mathrm{mean}((a-1)^2)$: L2 on normalized weights cannot collapse the distance to zero.

Quadratic is pulled toward a linear filter; Mahalanobis toward isotropic geometry.

GitHub snapshot, 9 September 2026: GrHa/06d and GrHa-learnable-pooling/06b. Geometry coefficient is provisional; GrHa/06b ReLU + mean uses only output L2.

Evaluate phenotype association and stability

Association across test donors

$$\mathrm{AUC}_{\mathrm{frequency}} = \mathrm{ROC\text{-}AUC}(y_b, f_b)$$

$$\Delta f = \overline{f}_{\mathrm{CMV}+} - \overline{f}_{\mathrm{CMV}-}$$

These complement network AUC. Do not reverse the direction after observing a test AUC below 0.5.

Each donor is an observation; additional bags do not create independent donors.

Useful additional checks

- Similar marker profiles across donor splits and random seeds.
- Actual marker coexpression in high-response cells.
- Sensitivity to extreme training responses that set the half-maximum threshold.

Frequency AUC measures association; stability and biological plausibility help assess whether the population is reproducible.

Project metrics: GrHa/06d. Stability and threshold-sensitivity checks are methodological recommendations, not reported results.

Four cell-filter models, one evaluation protocol

20 donors · 37 markers · `gated_alive`

100 shared outer splits: 14 training / 6 test donors.

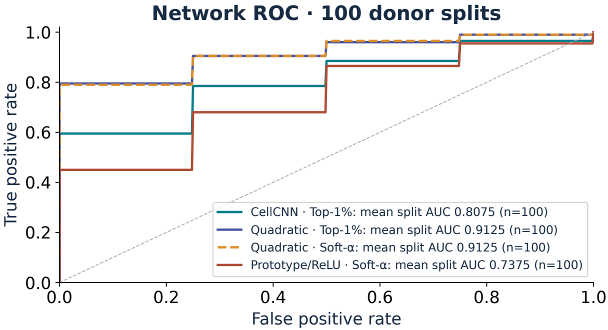
Model	Cell response $h_k(x)$	Pooling
CellCNN	$\mathrm{ReLU}(b_k + w_k^\top x)$	hard Top-1 %
Quadratic Top-1 %	$\mathrm{ReLU}(b_k + w_k^\top x + \sum_j q_{kj} x_j^2)$	hard Top-1 %
Quadratic Soft-α	$\mathrm{ReLU}(b_k + w_k^\top x + \sum_j q_{kj} x_j^2)$	learnable Soft-α
Prototype Soft-α	$\mathrm{ReLU}(\rho_k - \frac{1}{d} \sum_j a_{kj}(x_j - c_{kj})^2)$	learnable Soft-α

Population: largest positive output contrast; cells with $h > \frac{1}{2} \max_{\text{inner-train}} h$.

Donor prediction and CMV-status association of the selected population are evaluated separately.

Cohort: 9 CMV+, 11 CMV−. Each split: train 7+/7−, test 2+/4−. ArcSinh(x/5); scaling and half-max reference use the selected model's inner-training donors only.

Network classification across 100 donor splits

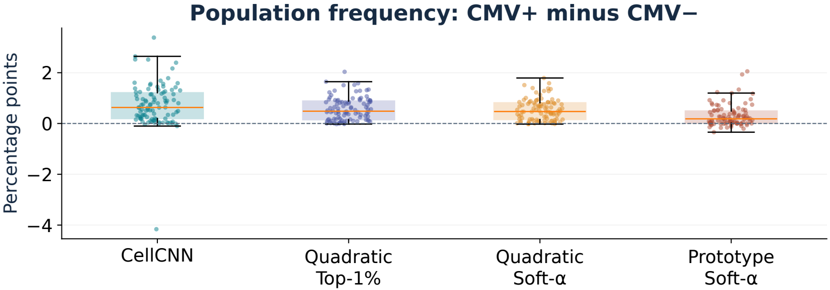


Model	Mean	Median
CellCNN	0.8075	0.8750
Quad. Top-1%	0.9125	1.0000
Quad. Soft-α	0.9125	1.0000
Prototype	0.7375	0.7500

Quadratic Top-1% and Soft-α share the highest mean network AUC (0.9125).

100 outer test sets, each with 2 CMV+ and 4 CMV− donors. Donor score: mean of five prediction bags of 20,000 cells. Curves are averaged per split; legend values are mean original split AUCs.

Selected-population frequency difference

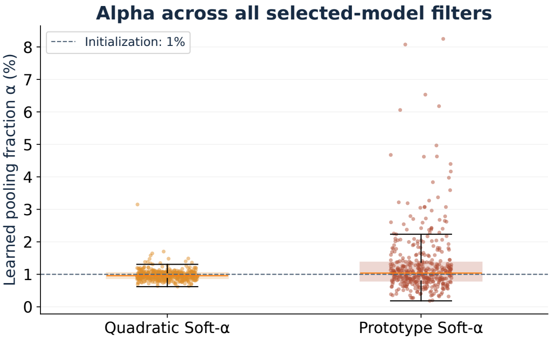


98 jointly evaluable splits. Boxes: Q1–Q3; line: median; dots: splits.

Largest mean frequency difference: CellCNN (0.740 percentage points).

98 jointly evaluable splits; test set: 2 CMV+ and 4 CMV− donors. One fixed 20,000-cell sample per test donor. Effect = mean CMV+ frequency minus mean CMV− frequency, in percentage points.

What does the extra pooling parameter learn?



Rank fraction, not population frequency

$$\alpha_k = \sigma(\theta_k)$$

Starts at 1%; fixed temperature $\tau = 0.002$.

Alpha weights cell ranks within a bag. Half-max defines the separately evaluated population.

Median α: Quadratic 0.96%; Prototype 1.04%. Alpha controls pooling, not measured population frequency.

All filters from 100 selected models per variant: 416 Quadratic and 414 Prototype filters. These are model parameters, not independent donor observations.

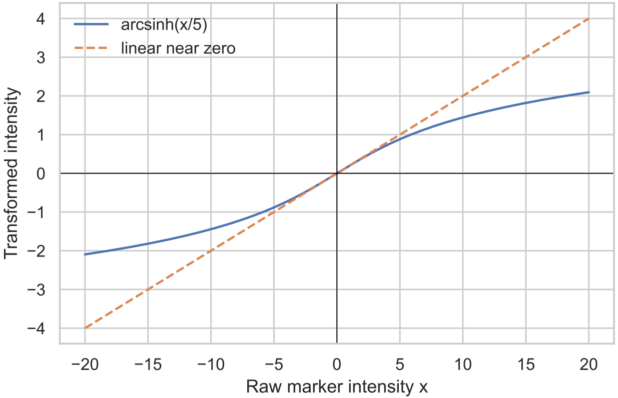
Take-away

- Quadratic Hard and Soft share the highest mean network AUC on this dataset.
- Soft vs. Hard: 5 better, 90 tied, 5 worse; mean ΔAUC 0.0000.
- Quadratic Soft-α vs. CellCNN: 53 better, 37 tied, 10 worse; mean ΔAUC $+0.1050$.
- Only 20 independent donors: repeated splits are descriptive, not independent replication.

Quadratic Hard and Soft both reach 0.9125 mean network AUC; learnable soft pooling does not improve it here.

All 100 planned outer splits are included. Paired differences use the same test donors in each split. These results do not establish performance on a new cohort.

Backup: Data Transformation and z-scoring



The arcsinh transformation

$$a_{ij} = \operatorname{asinh}(x_{ij}/5)$$

Approximately linear near zero; compresses large magnitudes and accepts negative background values.

Marker-wise z-scoring

$$z_{ij} = \frac{a_{ij} - \bar{a}_j}{s_j}$$

Prevents large marker scales from dominating distances. All 37 markers pass the variance (0.05) filter.

cofactor 5; pooled scaling across the balanced exploratory sample. Curve is a mathematical illustration.

Backup: how t-SNE turns neighbours into a map

High-D affinity (probability that j is a neighbour of i):

$$p_{ij} \propto \exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)$$

Low-D affinity (heavy-tailed, avoids "crowding"):

$$q_{ij} \propto (1 + \|y_i - y_j\|^2)^{-1}$$

Optimisation moves points in 2D until q_{ij} matches p_{ij} as closely as possible (minimises $\mathrm{KL}(P\|Q)$).

σ_i is tuned per point so its affinities imply the chosen perplexity.

perplexity	swept (5–100); effective neighbourhood size – see main slide.
init = pca	starts from the PCA layout instead of random noise: more reproducible, avoids poor local optima.
learning rate = auto	optimiser step size; scikit-learn scales it with sample size.
early exagg. = 12	temporarily inflates high-D affinities early on, so initial clusters separate before fine positioning.
angle = 0.5	Barnes–Hut speed/accuracy trade-off; a numerical setting, not a property of the data.
max_iter = 750	fixed iteration budget – not itself a convergence check (see caveat below).

van der Maaten & Hinton (2008); scikit-learn TSNE documentation. A fixed iteration count alone does not confirm convergence. Defaults refer to the documented scikit-learn 1.9 environment.

Backup: kNN preservation (R)

Let $N_i^X(k)$, $N_i^Y(k)$ be the k nearest neighbours of cell i in the high-D reference X and the embedding Y (excluding itself).

$$R(k) = \frac{1}{n} \sum_{i=1}^{n} \frac{|N_i^X(k) \cap N_i^Y(k)|}{k}$$

Symmetric; counts exact shared identities, not just "close enough."

What k controls. How many neighbours count as "local." Small k is a strict test of fine-grained ordering; large k is a looser test that tolerates more reshuffling and starts to resemble cluster-membership agreement rather than precise neighbour identity.

Why $k = 15$ here. Fixed throughout to match the neighbourhood scale explored in the UMAP sweep (`n_neighbors` 5–50) and sits within the commonly used 10–30 range for trustworthiness-style scores at this dataset size.

Example: ten shared neighbours out of fifteen contribute $10/15$.

Venna & Kaski (2001); scikit-learn NearestNeighbors documentation. Full-sample scores average over cells; equal cells per donor give equal donor weight.

Backup: Trustworthiness (T)

Using the same $k = 15$ and the same neighbour sets $N_i^X(k)$, $N_i^Y(k)$ as R:

$$T(k) = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_i \sum_{j \in N_i^Y(k) \setminus N_i^X(k)} (r_X(i,j) - k)$$

Penalises *false* low-D neighbours (points that look close in the embedding but weren't close in the reference) according to how far out of the true top-k they actually ranked. Unlike R, direction matters here: T only punishes false positives, not missed true neighbours.

Example: a false neighbour originally ranked 16th (just outside the true top-15) contributes only $(16 - 15) = 1$ to the penalty; one ranked 500th contributes 485 – a much larger embedding error is penalised much more heavily.

Venna & Kaski (2001); scikit-learn trustworthiness documentation. Here $k = 15$, matching R.

Backup: how UMAP constructs and embeds a weighted graph

$$w_{j|i} = \exp\left[-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right], \qquad w_{ij} = w_{j|i} + w_{i|j} - w_{j|i}w_{i|j}.$$

Offset ρ_i and scale σ_i adapt to local density; directed neighbourhoods are combined into symmetric weights.

$$v_{ij} = (1 + a\|y_i - y_j\|^{2b})^{-1}, \qquad L \sim -\sum_{i<j}\{w_{ij}\log v_{ij} + (1 - w_{ij})\log(1 - v_{ij})\}.$$

Sampled updates balance attraction and repulsion. The parameters a, b depend on min_dist and spread.

Swept	n_neighbors: 5, 15, 30, 50; min_dist: 0, 0.1, 0.3, 0.5.
Fixed/default	2 dimensions, seed 42, Euclidean metric, spectral initialisation, spread 1.

Larger neighbourhoods change the scale of the graph; larger min_dist discourages tight packing. Neither guarantees accurate global distances.

McInnes, Healy & Melville (2018/2020); official UMAP parameter and algorithm documentation. Equations describe the conceptual objective, not every optimisation approximation.

Backup: how UMAP constructs and embeds a weighted graph

Step 1 – build the high-D graph.

$$w_{j|i} = \exp\left[-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right], \qquad w_{ij} = w_{j|i} + w_{i|j} - w_{j|i}w_{i|j}.$$

Offset ρ_i and scale σ_i **adapt to local density**; directed neighbourhoods are combined into symmetric edge weights.

Step 2 – fit a low-D layout to match it.

$$v_{ij} = (1 + a\|y_i - y_j\|^{2b})^{-1}, \qquad L \sim -\sum_{i<j}\{w_{ij}\log v_{ij} + (1 - w_{ij})\log(1 - v_{ij})\}.$$

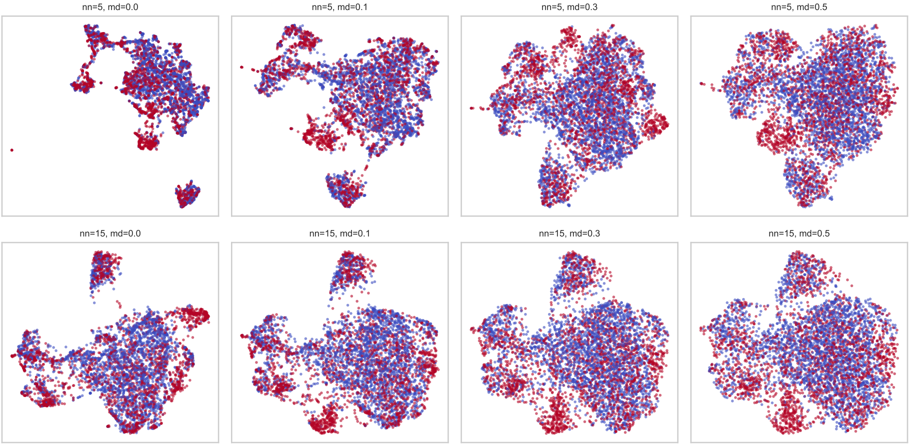
Sampled updates balance **attraction** (true neighbours pulled together) and **repulsion** (non-neighbours pushed apart). a, b are fit automatically from `min_dist`/`spread`, not set directly.

Swept	**n_neighbors**: 5, 15, 30, 50; **min_dist**: 0, 0.1, 0.3, 0.5.
Fixed/default	2 dimensions, seed 42, Euclidean metric, spectral initialisation, spread 1.

Larger neighbourhoods change the **scale** of the graph; larger `min_dist` discourages tight packing. Neither guarantees accurate global distances.

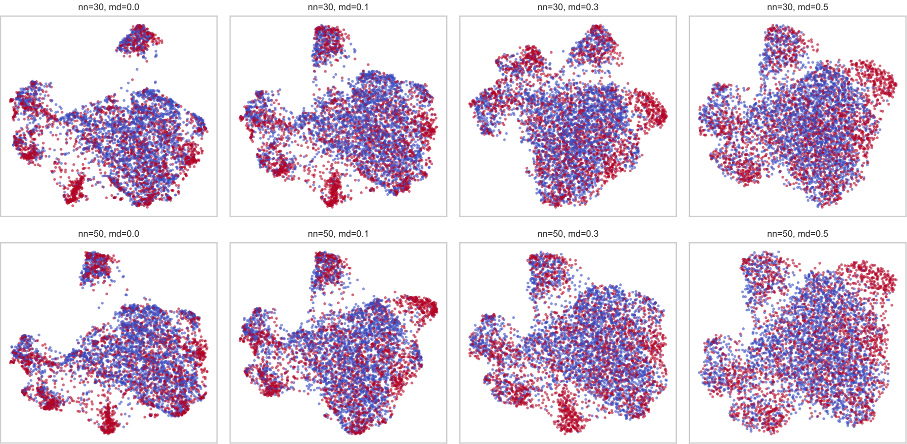
McInnes, Healy & Melville (2018/2020); official UMAP parameter and algorithm documentation. Equations describe the conceptual objective, not every optimisation approximation.

Backup: UMAP sweep, 5 and 15 neighbours



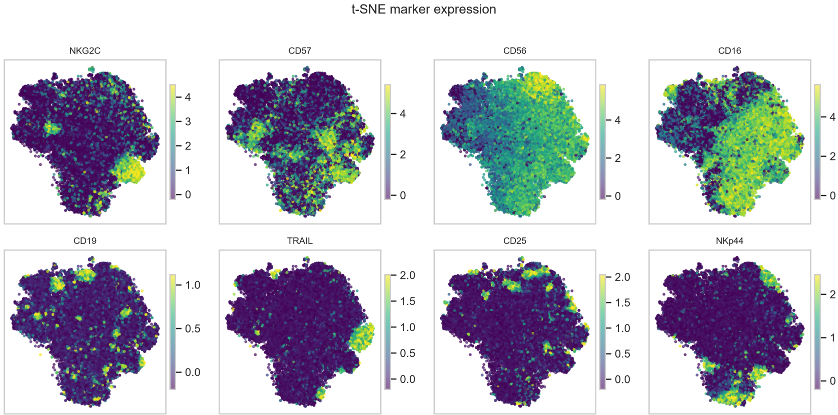
Rows 1–2 of the original 16-panel grid. Same 5,000 cells, 29 PCs, seed 42; Visual changes are parameter effects within one run, not a stability study across seeds.

Backup: UMAP sweep, 30 and 50 neighbours



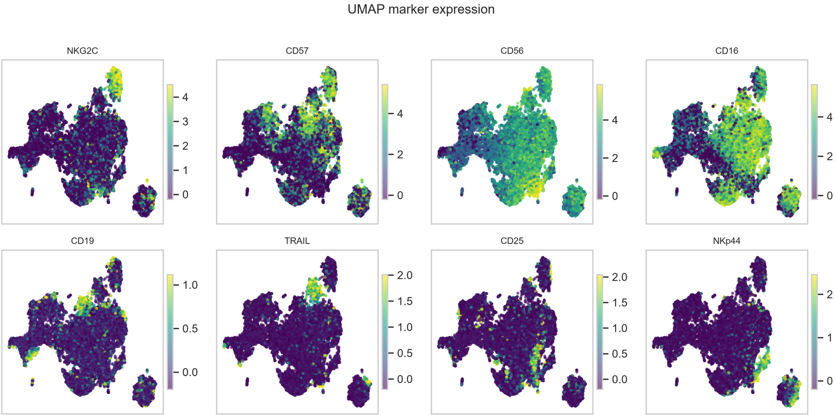
Rows 3–4 of the original 16-panel grid. Same 5,000 cells, 29 PCs, seed 42; Visual changes are parameter effects within one run, not a stability study across seeds.

Backup: the same markers on the t-SNE map



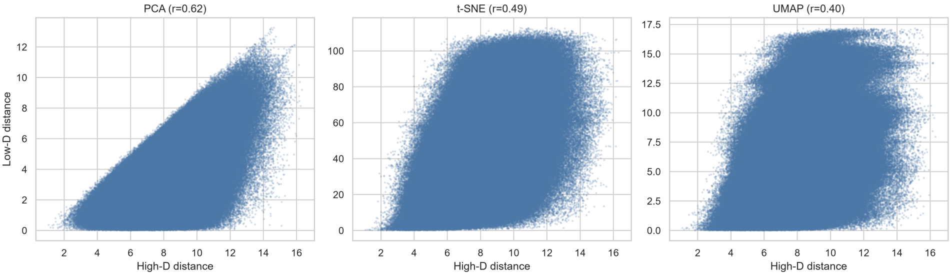
40,000 balanced cells. Arcsinh-transformed marker values; colour limits use the 1st/99th percentiles separately per marker. A bright region does not establish an infected-cell population.

Backup: marker expression on the UMAP map



40,000 balanced cells. Arcsinh-transformed marker values; colour limits use the 1st/99th percentiles separately per marker. A bright region does not establish an infected-cell population.

Backup: Shepard diagram to compare distances



Each point represents one sampled cell pair. The **horizontal axis** is the distance in the 29-dimensional PCA reference space.

The **vertical axis** is the distance in the two-dimensional embedding. Pearson correlation measures agreement in relative distances.

UMAP's loss doesn't optimize for preserving absolute distances at all, and `min_dist`=0 actively encourages tight packing, so many far-apart high-D pairs get compressed into a similar low-D distance ceiling $\rightarrow$ lowest r_d of the three

Fixed 2,000-cell sample; 29-PC reference space; Pearson correlation of pairwise distances.

Backup: Procrustes aligns the maps before measuring mismatch

Centre and normalise each coordinate matrix to unit Frobenius norm:

$$A_c = \frac{A - \bar{A}}{\|A - \bar{A}\|_F}, \qquad B_c = \frac{B - \bar{B}}{\|B - \bar{B}\|_F}.$$

$$M^2 = \min_{s, Q:\, Q^\top Q = I} \|A_c - sB_c Q\|_F^2.$$

- Rows must correspond to exactly the same cells in the same order.
- Rotation/reflection, translation and uniform scaling do not count as biological differences.
- Lower disparity means a closer global fit after alignment; it does not establish local fidelity or class separation.
- Nonlinear deformations can change neighbour identities and global shape differently.

SciPy scipy.spatial.procrustes. With its normalisation and optimal scaling, the scalar disparity is symmetric even though the returned transformed matrices have different normalisation conventions.

Backup: Procrustes aligns the maps before measuring mismatch

Centre and normalise each coordinate matrix to unit Frobenius norm:

$$A_c = \frac{A - \bar{A}}{\|A - \bar{A}\|_F}, \qquad B_c = \frac{B - \bar{B}}{\|B - \bar{B}\|_F}.$$

$\|M\|_F = \sqrt{\sum_{ij} M_{ij}^2}$ – treat the matrix as one long list of coordinates and take its ordinary (Euclidean) length; this just fixes overall "size" so only shape, not scale, is being compared.

$$M^2 = \min_{s,Q:\, Q^\top Q = I} \|A_c - sB_c Q\|_F^2.$$

- Rows must correspond to exactly the same cells in the same order.
- Rotation/reflection, translation and uniform scaling do not count as biological differences.
- Lower disparity means a closer global fit after alignment; it does **not** establish local fidelity (whether each point's *immediate* neighbours are correctly placed – that's what T/R measure instead) or class separation.
- Nonlinear deformations can change neighbour identities and global shape differently.

SciPy scipy.spatial.procrustes. With its normalisation and optimal scaling, the scalar disparity is symmetric even though the returned transformed matrices have different normalisation conventions.

Backup: Leiden 1/2: the graph and the objective

Build a graph from marker profiles

- Each cell becomes a node.
- Find neighbouring cells in the 37-dimensional marker space.
- Weighted edges represent local similarity between cells.

Neighbour count controls the scale of the graph. Our selected graph uses 30 neighbours.

Find communities in that graph

Leiden searches for a partition with a high graph-quality score.

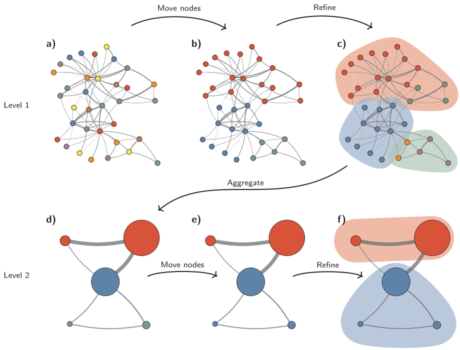
In our setup, the score rewards more internal connection weight than expected from node connectivity.

Resolution controls the penalty for the expected connections. Higher values usually yield smaller communities.

We set graph and resolution parameters; the number of clusters emerges. Here: resolution 1.2, 18 groups.

Traag, Waltman & van Eck (2019), pp. 1–2; our notebook uses Scanpy / leidenalg. Graph construction precedes Leiden. The selected objective is the configuration-model (RB) objective, not CPM.

Backup: Leiden 2/2: move, refine and aggregate



1. Move nodes (a–b)
Start with one group per node. Move nodes when this improves the quality score.

2. Refine groups (b–c)
Within each group, rebuild connected subgroups. Weakly joined parts can stay separate.

3. Aggregate (c–d)
Each refined subgroup becomes a node. Repeat on the smaller graph (d–f).

Refinement keeps subgroups available for later moves, instead of permanently collapsing an entire group. Improvement from Louvain.

Figure 3, cropped: Traag, Waltman & van Eck (2019), From Louvain to Leiden, CC BY 4.0.

Backup: K-means initialisation and Ward merges

K-means++ and repeated starts

- Standard k-means++ samples centres in proportion to squared distance from existing centres.
- scikit-learn uses a greedy variant with several candidates at each step.
- Our `n_init=10` retains the run with the lowest inertia.
- A fixed seed reproduces a run. Our resampling also changes the seed.

Ward minimises the increase in SSE

$$\Delta(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2$$

Ward prefers merges that add little within-cluster variation.

n_A, n_B: numbers of cells in groups A, B. μ_A, μ_B: their mean vectors. Δ: added squared error.

Ward only merges. K-means can reassign cells.

scikit-learn KMeans and SciPy linkage documentation. SSE = sum of squared errors.

Backup: geometry and the singleton example

Silhouette

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$a(i)$: mean distance to other cells in its cluster.
$b(i)$: smallest mean distance to another cluster.

Resolving a nearby subgroup can reduce silhouette even if the subgroup is biologically meaningful. This is a possibility, not an observed comparison in our run.

Davies–Bouldin with cluster and singleton

$$S_c = \frac{1}{n_c} \sum_{x_i \in C_c} \|x_i - \mu_c\|$$

$$DB = \frac{S_{\text{large}} + 0}{\|\mu_{\text{large}} - x_{\text{singleton}}\|}$$

C_c: cluster c; n_c: its cell count. x_i: a cell vector; μ_c: the cluster mean.
A distant singleton has zero scatter and can yield a low DB.

scikit-learn silhouette and Davies–Bouldin definitions. No claim that silhouette always selects too few clusters, or that DB is neutral across cluster shapes.

Backup: sensitivity to the stability cutoff

Mean stability cutoff	0.50	0.60	0.65	0.70	0.75
K-means: selected clusters	4	4	4	4	4
Leiden: selected clusters	18	18	18	7	6
Ward: selected clusters	6	6	2	—	—

The cutoff affects the chosen resolution.

K-means remains at four clusters over these tested thresholds.

Stricter thresholds select coarser Leiden solutions.

"—" means no eligible configuration. Single linkage remains at $k = 2$ throughout.

The 0.6 threshold is a transparent heuristic. Sensitivity analysis shows which conclusions depend on it.

`notebooks/03_clustering.ipynb`, 7 September 2026. Initial seed 42; 10 K-means starts; 44 reference and 440 subsampling fits.
$k \in \{2, 3, 4, 6, 8, 10, 12\}$; neighbours $\in \{10, 30, 60\}$; resolutions $\in \{0.3, 0.7, 1.2\}$.

Backup: from K-means loss to Gaussian likelihood

$$
\begin{aligned}
(r, m) &= \arg\min_{r,m} \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \|x_i - m_k\|^2 \\
&= \arg\max_{r,m} \left[\sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \left(-\frac{\|x_i - m_k\|^2}{2\sigma^2} \right) + \mathrm{const.} \right] \\
&= \arg\max_{r,m} \prod_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \frac{\exp\left(-\|x_i - m_k\|^2/(2\sigma^2)\right)}{Z} \\
&= \arg\max_{r,m} \prod_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \mathcal{N}(x_i; m_k, \sigma^2 I_d) \\
&= \arg\max_{r,m} p(X \mid m, r).
\end{aligned}
$$

Adapted from Macke / Hennig, Probabilistic Machine Learning (2024), Lecture 11, slide 15, . Binary one-hot r; fixed shared $\sigma^2 > 0$; $Z = (2\pi\sigma^2)^{d/2}$. Component index corrected to m_k.

Backup: hyperparameters and paper deviations

	Implemented settings	Selection / paper relationship
CellCNN	3,000 cells/input; 200 inputs/donor; 3/4/5 filters; top 1%; Adam 0.01; batch 128; L2 10^{-4}; no dropout; $\leq$ 100 epochs; patience 5.	Best validation accuracy; ties: AUC, loss, fewer filters, fold order. Fixed search; five rather than one test input. Selected net uses 9/10 donors.
Citrus	10,000 cells/donor; Ward; 37 ArcSinh markers; minimum cluster fraction 0.0005; L1 logistic; cv.min; no marker z-scoring.	10,000 cells/30 repeats versus paper's 20,000/100. Inner trees fitted per fold; original lambda grid uses all 14 outer training donors. Final fit on 14 donors.
SVM	10,000 cells/training donor; train-only scaler; L2 squared hinge; loss weight $C \in \{0.01, 0.1, 1\}$; top 1% of full-test margins (count rounded up).	Maximize mean inner AUC; ties: smaller C. Inner OOF threshold maximizes sensitivity + specificity. Final fit on 14 donors. Top-1-% aggregation is our adaptation.

Arvaniti & Claassen (2017), doi:10.1038/ncomms14825. Bruggner et al. (2014), doi:10.1073/pnas.1408792111.

Backup: SVM approach 1 — training the cell scorer

Training data and scaling

Sample 10,000 cells per training donor without replacement, with fixed seeds. Each cell inherits its donor's label. Fit a shared scaler on these training cells only.

$$a_{ij} = \operatorname{asinh}(x_{ij}/5), \qquad z_{ij} = \frac{a_{ij} - \mu_j}{\sigma_j}$$

$\boldsymbol{i}$: cell; $\boldsymbol{j}$: marker; $\boldsymbol{x_{ij}}$: raw value; $\boldsymbol{a_{ij}}$: transformed value; 5: cofactor; $\boldsymbol{\mu_j, \sigma_j}$: training mean and standard deviation; $\boldsymbol{z_{ij}}$: standardized value.

Equal cell counts give donors equal weight in the training loss. No extra class weighting.

Labels provide weak cell supervision. The 37 markers yield 37 learned weights and one intercept.

Linear score and squared-hinge loss

$$m_i = \sum_{j=1}^{37} w_j z_{ij} + b$$

$$\min_{w,b} \left[\tfrac{1}{2}(\|w\|^2 + b^2) + C \sum_{i=1}^{N} \max(0, 1 - y_i m_i)^2\right]$$

$\boldsymbol{w}$: vector of marker weights w_j; $\boldsymbol{b}$: intercept; $\boldsymbol{m_i}$: cell margin; $\boldsymbol{N}$: training-cell count; $\boldsymbol{y_i}$: donor label, -1 for CMV$-$ and $+1$ for CMV$+$.

$\boldsymbol{C}$: loss weight; $\|\boldsymbol{w}\|^2$: sum of squared weights; $\mathbf{min}_{\boldsymbol{w},\boldsymbol{b}}$: optimize weights and intercept. The squared maximum is the squared-hinge loss.

Smaller $\boldsymbol{C}$: more regularization. Larger $\boldsymbol{C}$: more weight on fitting labels.

`LinearSVC` with `intercept_scaling=1` also penalizes b. No probability calibration.

Backup: SVM approach 2 — model selection and test

1. Choose settings within 14 donors

Use 3 inner donor folds. Test loss weights $C \in \{0.01, 0.1, 1\}$ on identical cell samples and scalers within each fold.

Highest mean inner donor ROC-AUC wins; ties favour smaller C. For that C, collect one out-of-fold score per training donor.

$$J(\tau) = \mathrm{sens}(\tau) + \mathrm{spec}(\tau) - 1$$

$\boldsymbol{\tau}$: candidate donor threshold; $\boldsymbol{J}$: Youden index; **sens**: fraction of CMV+ donors predicted positive; **spec**: fraction of CMV− donors predicted negative.

Maximize J over finite ROC thresholds; ties choose the highest offered optimum.

2. Refit, then evaluate six test donors

Resample 10,000 cells from each of the 14 training donors; fit a new scaler and SVM. Keep the inner-selected threshold.

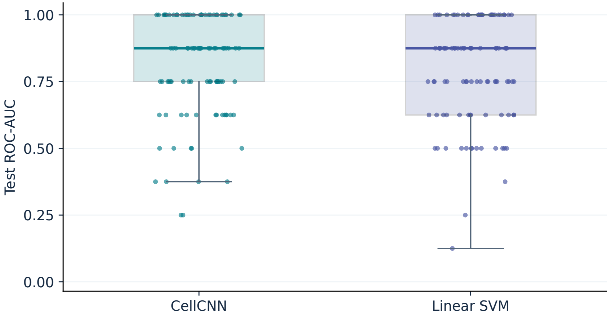
$$k_d = \lceil 0.01 N_d \rceil, \qquad s_d = \frac{1}{k_d} \sum_{i \in I_d} m_i$$

$\boldsymbol{d}$: donor; $\boldsymbol{N_d}$: its full cell count; $\boldsymbol{k_d}$: count rounded up; $\boldsymbol{I_d}$: top-scoring cell indices; $\boldsymbol{i}$: cell; $\boldsymbol{m_i}$: margin; $\boldsymbol{s_d}$: donor score.

Predict CMV+ when $s_d \geq \tau$. AUC uses continuous scores; balanced accuracy uses thresholded labels.

The uncalibrated threshold transfers from inner models to a refitted model. Changes in score scale can affect thresholded predictions. Outer test donors choose neither C nor τ.

Backup: CellCNN and SVM across 100 splits



Separate two-method comparison

Method	AUC	AP	BA
CellCNN	0.875	0.833	0.750
Linear SVM	0.875	0.833	0.625

Medians across splits 0–99.
AUC: ROC-AUC; AP: average precision;
BA: balanced accuracy.

600 held-out donor predictions per method, from the same 20 donors.

Both median AUCs remain 0.875. More splits do not add independent donors.

Boxes: Q1–Q3; dots: individual splits; dashed line: AUC 0.5.

Backup: CellCNN subset selection in detail

All quantities refer to one saved model and its original training reference.

Response and reference maximum

$$r_{if} = \max(0, w_f^\top z_i + b_f)$$

$$M_f = \max_{\ell \in T} r_{\ell f}, \qquad q_{if} = r_{if} / M_f$$

$\boldsymbol{i}, \boldsymbol{\ell}$: cells; $\boldsymbol{f}$: filter; $\boldsymbol{z_i}$: 37 markers standardized by the saved scaler; $\boldsymbol{w_f}$: weights; $\boldsymbol{b_f}$: bias; $\boldsymbol{r_{if}}$: ReLU response; $\boldsymbol{w_f^\top z_i}$: weighted sum.

$\boldsymbol{T}$: reference cells; $\boldsymbol{M_f}$: maximum response; $\boldsymbol{q_{if}}$: normalized score, defined for $M_f > 0$.

Reconstruct **20,000 cells per actual training donor** with the original seed: 9 or 10 donors, hence 180,000 or 200,000 reference cells.

Direction, subset and centroid

$$\Delta_f = v_{f,+} - v_{f,-}$$

$$S_f = \{i \in T : q_{if} > 0.5\}, \qquad c_f = \frac{1}{|S_f|} \sum_{i \in S_f} a_i$$

$\boldsymbol{v_{f,+}}, \boldsymbol{v_{f,-}}$: weights from filter f to the CMV+/CMV− outputs; $\boldsymbol{\Delta_f}$: output contrast. Its sign gives the direction.

$\boldsymbol{S_f}$: selected reference cells; $|\boldsymbol{S_f}|$: their count; $\boldsymbol{a_i}$: ArcSinh marker vector; $\boldsymbol{c_f}$: mean marker vector (centroid).

Skip filters with zero maximum or zero output contrast. Equality at half-maximum is excluded.

Selected cells have equal weight; donor contributions vary. New cells can score above 1.

Backup: Citrus subset selection in detail

Which frequency features matter?

$$\eta_d = \beta_0 + \sum_j \beta_j h_{dj}$$

$\boldsymbol{d}$: donor; $\boldsymbol{j}$: cluster; $\boldsymbol{h_{dj}}$: cluster-frequency feature; $\boldsymbol{\beta_0}$: intercept; $\boldsymbol{\beta_j}$: coefficient; $\boldsymbol{\eta_d}$: linear predictor for CMV+.

`glmnet` standardizes frequency features internally.

$$q_j = |\beta_j| > 10^{-10}$$

$\boldsymbol{q_j}$: absolute coefficient; $\mathbf{10^{-10}}$: numerical zero tolerance.

Retain every effective cluster. A positive coefficient raises the CMV+ predictor as frequency increases, with other features fixed; a negative coefficient lowers it.

What the exports preserve

$$c_j = \frac{1}{|A_j|} \sum_{i \in A_j} a_i$$

$\boldsymbol{A_j}$: training cells belonging to cluster j; $|\boldsymbol{A_j}|$: their count; $\boldsymbol{i}$: cell; $\boldsymbol{a_i}$: ArcSinh marker vector; $\boldsymbol{c_j}$: saved cluster centroid.

Training uses 10,000 cells/donor and retains clusters from 0.05% of training cells. Parent and child clusters can share cells.

Stored centroids and coefficients do not preserve the full tree and cell memberships needed to assign arbitrary map cells.

Assigning cells to the nearest centroid would introduce a different selection rule. Null models remain in the 30-split recurrence denominator.

Backup: SVM cell selection and recurrence

Exact selection within a test donor

$$k_d = \lceil 0.01 N_d \rceil, \qquad t_{is} = m_{is} - \tau_s$$

$\boldsymbol{d}$: donor; $\boldsymbol{s}$: outer split; $\boldsymbol{i}$: cell; $\boldsymbol{N_d}$: full cell count; $\boldsymbol{k_d}$: count rounded up; $\boldsymbol{m_{is}}$: cell margin; $\boldsymbol{\tau_s}$: learned donor threshold; $\boldsymbol{t_{is}}$: centered score.

Let $\boldsymbol{I_{ds}}$ contain the k_d highest-scoring cell indices. Break boundary ties by original FCS event number.

Within $\boldsymbol{I_{ds}}$: positive if $t_{is} > 0$, negative if $t_{is} < 0$, neither if $t_{is} = 0$. Cells outside I_{ds} are unselected.

Example: 101 cells give 2 top cells. Margins 9 and 8 at threshold 8.5 give one positive and one negative selection.

Repeat across that donor's test visits

$$f_i^+ = \frac{n_i^+}{|\mathcal{T}_d|}$$

$\boldsymbol{\mathcal{T}_d}$: all splits testing donor d; $|\boldsymbol{\mathcal{T}_d}|$: their count; $\boldsymbol{n_i^+}$: visits where cell i is in I_{ds} and $t_{is} > 0$; $\boldsymbol{f_i^+}$: positive selection frequency.

Include visits without selection: 3 positive selections in 8 test visits give 3/8.

$$s_{ds} - \tau_s = \frac{1}{k_d} \sum_{i \in I_{ds}} t_{is}$$

$\boldsymbol{s_{ds}}$: mean of the selected cell margins, the donor score in split s.

Fixed-selection decomposition of the donor score; no causal cell-removal effect.

Backup: CellCNN — from cells to one plotted point

Slide 30: one selected model per split, one subset profile per effective filter

1. Reconstruct the training reference
20,000 cells per actual inner training donor; 9 or 10 donors give 180,000 or 200,000 reference cells. Reuse the original seed.

2. Select cells for each filter
Apply the saved scaler and filter. Keep responses strictly above half of that filter's reference maximum.

3. Average the selected marker profiles
Average each of the 37 ArcSinh-transformed markers across the selected cells. One filter gives one centroid, carrying its split and filter ID.

Actual example: split 9, filter 2

Reference cells	200,000
Half-maximum-selected cells	1,633
Result	1 centroid

This centroid becomes the representative of G1. The later cell map applies the same filter to a different, 10,000-cell display sample.

Mean, not median
Illustrative values for one marker: 1, 2, 9. Their mean is 4; their median is 2. We use the mean.

Across 30 selected models: 7 have 3 filters, 6 have 4, and 17 have 5. Together they yield **130 subset centroids**.

Backup: Citrus — two levels of clustering

Slide 30: original cell clusters first, groups of their profiles afterwards

1. Cluster cells within each split

The final Citrus model uses 10,000 cells from each of 14 training donors: **140,000 cells**.

A Ward tree defines cell clusters. L1-logistic regression uses their donor frequencies and retains effective coefficients of either sign.

For every selected cluster, average each ArcSinh marker over its training members. This gives **one centroid per selected cluster**.

Across 30 splits: **123 selected cluster centroids**. Three splits select no effective clusters; they remain in the recurrence denominator.

2. Group profiles across the splits

Compare the 123 centroids in the common standardized marker space. Similar profiles form **17 groups**; 5 occur in at least 6 splits and are shown.

Each coloured point still belongs to an original cluster in a specific split. A colour identifies its profile group.

Actual example: Citrus G1

28 cluster centroids from **18 splits**.

Its star is one existing member: cluster 139891 from split 25. It is not an average of all 28 cluster profiles.

Backup: Colours, stars and overlapping centroids

Slide 30: grouping and display are separate operations

Colour: a group of similar profiles
Use all 37 markers on the shared exploratory z-scale. Group separately for each method with cosine distance, average linkage and a cut at 0.4.

Average linkage averages pairwise distances between groups. It does not replace their members with one mean profile.

Coefficient signs are ignored during grouping: a group may be positive, negative or mixed. Show groups present in at least **6 distinct splits**; each split counts once.

Star: an existing representative
Choose the member with the smallest summed cosine distance to its group members. This is the **medoid**, not a median or a new mean profile.

Position: the nearest map cell
Reuse the Task 2 map: 10,000 cells, 500/donor. Assign each centroid the t-SNE position of its nearest map cell in standardized marker space (Euclidean distance).

Several centroids can share a position. The star need not look central in the 2D map.

	All centroids	Shown groups	Plotted centroids	Distinct positions
CellCNN	130	4 of 15	111	44
Citrus	123	5 of 17	77	52

Backup: Individual cells and SVM selection frequency

Slide 31: each coloured point is one original cell in the 10,000-cell map

Left: one CellCNN representative

Use the G1 representative: **split 9, filter 2**. Apply its saved scaler and filter to all 10,000 map cells from all 20 donors.

Keep the original half-maximum threshold from its training reference; do not recompute a maximum on the map.

87 map cells pass and are coloured turquoise. Colour indicates selection, not response strength or recurrence.

The centroid used 1,633 selected reference cells. The 87 points are a separate application to the map sample. This is exploratory and includes training donors.

Right: repeated SVM test evaluations

Each saved SVM scores only its outer test donors. Select the highest 1% of each **full donor's** margins; positive selection also requires exceeding the learned donor threshold.

Count positive selections of the same map cell across **all test appearances of its donor**: 4–16 appearances, including those without selection.

Example: 3 positive selections in 8 test appearances give a frequency of **0.375**. Plot that cell once with this colour.

93 map cells pass at least once. Zero frequency stays grey; brighter colours mean more consistent positive selection.

Backup: Which cells enter each heatmap row?

Slide 32: eight displayed markers; all profiles start on the ArcSinh scale

Row	Cells used	How the profile is formed
CellCNN G1	Split 9/filter 2: 1,633 of 200,000 reference cells; 10 inner training donors.	Mean markers of one G1 representative; not all G1 centroids or the 87 map cells.
Citrus G1	Split 25/cluster 139891: its members within 140,000 training cells from 14 donors.	Mean markers of one G1 representative cluster; not all 28 G1 centroids.
SVM positive OOF	Positively selected cells from full outer test donors in splits 0–29; 168 of 180 visits are nonempty.	Average cells within visits; nonempty visits within donors; then the 20 donors equally.

Positive OOF
Top-1-% cells above the learned threshold, scored only when their donor was held out. Both CMV classes can contribute.

Equal donor weights
Each donor contributes equally, regardless of cell count or test visits. Use full-donor selections, not just the 93 map cells.

Backup: Reading the shared marker scale

Slide 32: expression relative to a common reference, not classifier weights

Standardize each mean marker value

$$z_j = \frac{c_j - \mu_j}{\sigma_j}$$

$\boldsymbol{j}$: marker
$\boldsymbol{c_j}$: profile's mean ArcSinh marker value
$\boldsymbol{\mu_j}$: reference mean of that marker
$\boldsymbol{\sigma_j}$: reference standard deviation
$\boldsymbol{z_j}$: value shown in the heatmap

The shared reference

10,000 map cells, 500 per donor, including both CMV classes. Reuse the stored exploratory scaler from Task 2 for every heatmap row.

The model-specific scalers still govern cell selection; this shared scale is used for display.

Read the number and colour together

Value	Relative to the reference mean
-1	1 standard deviation below
0	Equal to the reference mean
$+1$	1 standard deviation above

Example: CellCNN NKG2C = 4.6

4.6 reference standard deviations above the reference mean; not 4.6 times the expression.

Red: above reference. Blue: below. These signs do not describe CMV class or model direction. Zero means reference-level expression.

Mean profiles can hide heterogeneous cells. These values are not cell-type gates.

Backup: Same array: hard top-k versus sigmoid selection

$$[0,3,0,8,1,6,0,2] \xrightarrow{\text{sort}} [8,6,3,2,1,0,0,0]$$

Select the top 25 %: $k = 2$ for eight cells.

Sorted responses	8	6	3	2	1	0	0	0
Hard selection	1	1	0	0	0	0	0	0
Sigmoid selection	1.000	.958	.042	.000	.000	.000	.000	.000

$$P_{\text{top-2}} = \frac{8+6}{2} = 7 \qquad P_{\text{soft}} = \frac{\sum_r m_r s_{(r)}}{\sum_r m_r} \approx 6.937$$

Both focus on the same strongest cells. The sigmoid replaces the abrupt cutoff with a small transition.

Illustration: $\alpha = .25$, $\tau = .02$ (model: .002). Masks are rounded; the soft mean uses unrounded values.

Backup: Activation, pooling and population thresholds

Quantity	Definition	Role
Activation boundary	$h_k = \mathrm{ReLU}(\rho_k - d_k^2)$ or $h_k = \mathrm{ReLU}(z_k)$	Defines positive responses; learned through filter parameters.
Rank fraction α_k	$m_{rk} = \mathrm{sigmoid}((\alpha_k - u_r)/\tau)$	Controls pooling breadth; learned per filter.
Population threshold t_k	$t_k = \frac{1}{2} M_k$ $M_k = \max_{\text{Inner-Train}} h_k(x)$	Heuristic population cutoff; calibrated after training.

Half-maximum is a separate evaluation heuristic. It is neither the learned rank fraction nor a threshold trained by backpropagation.

Half-maximum analysis: GrHa/06d and shared discussion. Target Mahalanobis-ReLU model: apply the same rule to h_k.

Backup: distance geometry and response

Learn the geometry

$$\|x - c\|^2 \quad \longrightarrow \quad (x - c)^\top M(x - c)$$

A diagonal M stretches the contours along marker axes.

With $a_j = 1$, our normalized distance reduces to the mean squared Euclidean distance.

Choose the response

$$g(x) = \exp\big(-d^2(x)/(2\ell^2)\big)$$

$$h(x) = \mathrm{ReLU}\big(\rho - d^2(x)\big)$$

Gaussian: smooth, positive tails.
ReLU: zero response outside a learned boundary.

Mahalanobis does not require ReLU. We choose ReLU for an explicit inactive region, not because Gaussian responses are inherently worse.

Own comparison of the stated functions. Gaussian geometry: GPML, Ch. 4; ReLU prototype response: GrHa/06b.

Backup: a gradient reaches the learned fraction

Within a fixed response ordering, let $S_k = \sum_r m_{rk}$.

$$\frac{\partial m_{rk}}{\partial \alpha_k} = \frac{m_{rk}(1 - m_{rk})}{\tau}$$

$$\boxed{\frac{\partial P_k}{\partial \alpha_k} = \frac{1}{\tau S_k} \sum_r m_{rk}(1 - m_{rk})(s_{(r)k} - P_k)}$$

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = \frac{\partial \mathcal{L}}{\partial P_k} \frac{\partial P_k}{\partial \alpha_k} \alpha_k (1 - \alpha_k)$$

The term $s_{(r)k} - P_k$ accounts for the changing denominator. The loss determines whether the fraction should increase or decrease.

The learning path exists, but gradients may vanish. Sorting and ReLU are differentiable almost everywhere, not everywhere smooth.

Own quotient-rule derivation. Gradients to responses: $\partial P_k / \partial s_{(r)k} = w_{rk}$ within a fixed ordering.

Backup: why parameterize the threshold with softplus?

$$\rho_k = \mathrm{softplus}(\eta_k) + \varepsilon = \log(1 + e^{\eta_k}) + \varepsilon > 0$$

- $\eta_k \in \mathbb{R}$ is an unconstrained parameter updated by the optimizer.
- Softplus maps it smoothly to a positive squared-distance threshold.
- ε keeps the threshold strictly above zero numerically.

$$\frac{\partial \rho_k}{\partial \eta_k} = \mathrm{sigmoid}(\eta_k)$$

This is a positivity constraint implemented through a smooth parameterization, not an additional threshold or pooling operation.

GrHa/06b, radii(). Softplus is distinct from the sigmoid rank mask used in Soft Top Fraction.

Backup: positive, normalized marker weights

Parameterization

$$u_{kj} = \mathrm{softplus}(\beta_{kj}) + \varepsilon$$

$$a_{kj} = \frac{u_{kj}}{\frac{1}{p}\sum_{\ell} u_{k\ell}}$$

$$a_{kj} > 0, \qquad \mathrm{mean}_j\, a_{kj} = 1$$

Effect

The common scale of the weights is fixed.

The denominator couples all marker weights.

Normalization is a parameter constraint; the additional geometry penalty favors $a_{kj} \approx 1$.

A diagonal metric still models a joint marker profile, even without explicit cross-terms or rotations.

GrHa-learnable-pooling/06b, marker_weights.

Backup: half-max population frequency

Filter selection and fixed training threshold

$$\Delta_k = V_{1k} - V_{0k}, \qquad k^* = \arg \max_{k:\Delta_k > 0} \Delta_k$$

$$M = \max_{x \in \text{inner-train cells}} h_{k^*}(x), \qquad t = \tfrac{1}{2} M$$

$$f_i = \frac{\#\{x \in \text{test sample } i : h_{k^*}(x) > t\}}{N_i}$$

One fixed 20,000-cell sample per test donor. The strict rule also applies when $M = 0$.

Two population metrics

$$\mathrm{AUC}_{\mathrm{freq}} = \mathrm{AUC}(y_i, f_i)$$

$$\mathrm{Effect} = \bar{f}_{\mathrm{CMV+}} - \bar{f}_{\mathrm{CMV-}}$$

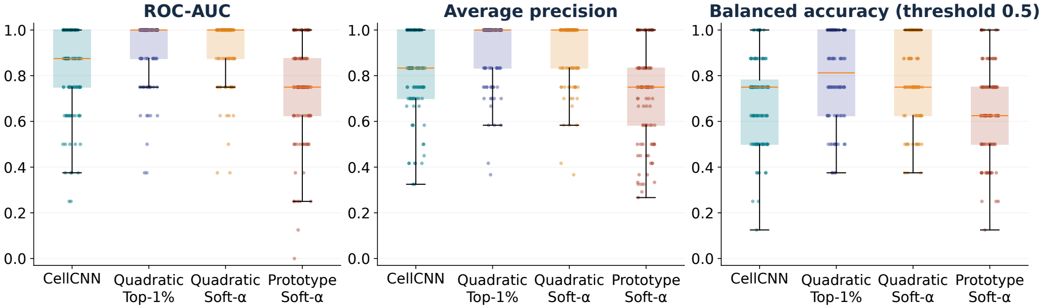
Without a positive filter, values remain missing. No post-hoc flipping of AUC direction.

Frequency comparisons use **98** jointly evaluable splits.

Half-max is cell selection, not clustering; CMV is a donor label, not a cell-type label.

The baseline has no positive-output filter in splits 44 and 84. These two splits are excluded from all common frequency comparisons, but retained for network metrics.

Backup: variation in classification across splits

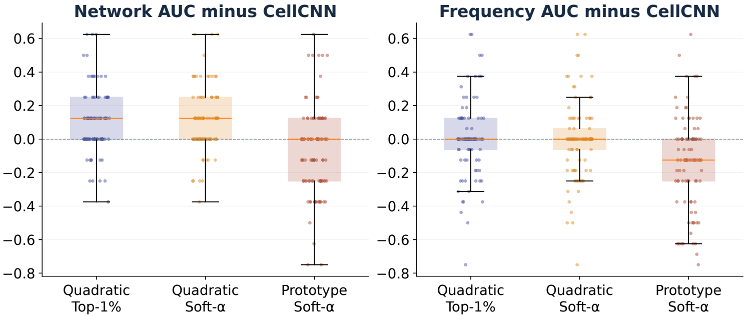


100 overlapping donor splits. Boxes: Q1–Q3; line: median; dots: splits. Not confidence intervals.

Highest means: Quadratic Top-1% for AP (0.907) and BA (0.801).

100 test sets of six donors (2 CMV+, 4 CMV−). AP: average precision; BA: balanced accuracy at score threshold 0.5. Boxes: Q1–Q3; line: median; dots: splits. Not confidence intervals.

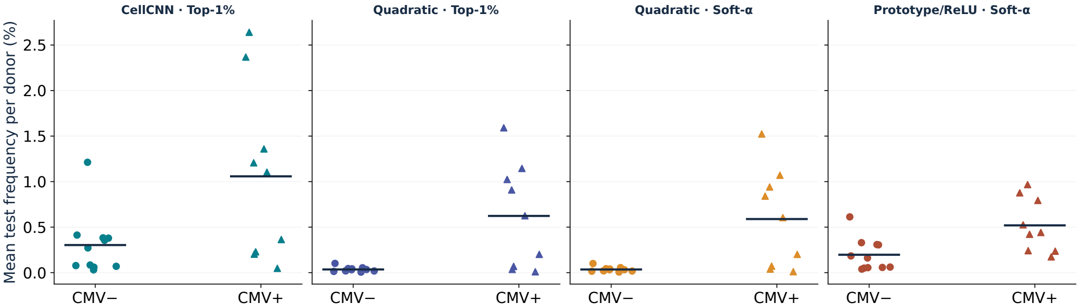
Backup: paired differences to CellCNN



Quadratic Soft-α vs. CellCNN: 53 better, 37 tied, 10 worse; mean ΔAUC $+0.1050$.

Network: 100 paired splits; frequency: 98 common paired splits. Difference = model AUC minus CellCNN AUC within the same split. Positive values indicate higher AUC, not independent replication.

Backup: population frequencies of individual donors

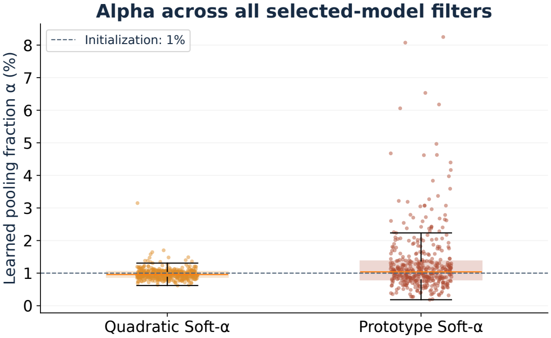


One point per donor: average across available test visits in the common splits. The selected population may vary by split.

Each donor contributes one point, averaging their available test visits first.

20 donors per model: 9 CMV+ (triangles), 11 CMV− (circles); visits restricted to the same 98 common splits. Bars: mean of donor points. The selected filter may differ between splits.

Backup: the learnable rank mask



Rank fraction, not population frequency

$$\alpha_k = \sigma(\theta_k), \qquad \alpha_k^{(0)} = 0.01$$

$$m_{rk} = \sigma\left(\frac{\alpha_k - q_r}{0.002}\right)$$

$q_r = (r - \frac{1}{2})/N$ for descending response ranks. Temperature is fixed. Soft $\alpha = 1\%$ is not identical to hard Top-1%.

Median α: Quadratic 0.96%; Prototype 1.04%. Alpha controls pooling, not measured population frequency.

100 selected models per variant; all 416 Quadratic and 414 Prototype filter parameters are shown. The temperature is not fitted or tuned in this comparison.

Backup: scope of the controlled comparison

- **Quadratic Hard vs. Soft:** same cell response and L2 penalty; the pooling function changes.
- **CellCNN vs. Quadratic:** adds a quadratic term and its L2 penalty.
- **Prototype vs. Quadratic:** different geometry, initialization and regularization; not an isolated alpha comparison.
- **No refit:** the selected inner model uses 9–10 training donors, not all 14 outer-training donors.
- **Limitation:** variants were developed after earlier results on these same 20 donors.

L2 coefficient 10^{-4}: CellCNN on W, V; Quadratic on W, Q, V; Prototype on V, plus $10^{-3}\,\mathrm{mean}((a-1)^2)$. No extra alpha or radius penalty.

Quadratic Hard and Soft both reach 0.9125 mean network AUC; learnable soft pooling does not improve it here.

Nine candidates per outer split: three inner folds and 3/4/5 filters. The same donor splits, sampling seeds, learning rate and early-stopping procedure are used.